Reg. No. | | | | | | | | | | |

[18MSCH3S31]

THIRD SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER - 2019

APPLIED CHEMISTRY

Time:03Hours **Maximum Marks:70**

Instructions:
1. Question **1** is compulsory, carries **10** marks.
2. Questions **2 to 7** carry **15** marks each. Answer any Four.

1. Answer ANY FIVE of the following: **5x2=10**

a. What are the effects of larger surface to volume ratio on material properties?

b. Calculate the rotational constant and bond length of CO from a rotational band line spacing of 3.86 cm^{-1}.

c. How modification of associated polymer results in the performance of contact lens?

d. Define chromophore with an example.

e. Describe Silver Button Battery.

f. Explain deactivation and regeneration of catalyst.

2a. Discuss Raman spectra of single, bi- and multi-layer (>50 layer) graphene in details with representative Raman spectra.

2b. Why core-shell quantum dots are important?. Describe in terms of their optical properties. **10+5**

3a. Describe in detail the electron energy loss spectroscopy in TEM, and also interaction of generated x-rays during the process with the material.

3b. Explain extended x-ray absorption technique and its applications. **10+5**

4a. Explain condensation polymerization with examples.

4b. i. What is glass transition temperature (Tg)?

ii. Explain the factors affecting on Tg.

iii. How Tg determined and give its significance?

4c. Describe the coordination polymerization reaction with the help of Zeigler- Natta catalyst. **5+5+5**

5a. Discuss the pigments, surfactants and thinner in details with suitable examples.

5b. Describe the synthesis and application of crystal violet dye. **10+5**

6a. Derive Nernst equation.

6b. Write a note on Nickel-Cadmium battery. Briefly describe its working principle, applications, merits and demerits.

6c. Write a note on lithium iodine primary battery **5+5+5**

7a. Derive the Langmuir-Hinshelwood mechanism.

7b. What are the advantages of sol gel method of catalyst preparation? **10+5**

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